THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS FULL SYLLABUS PAPER SET 05

Full Syllabus (Chapters 1–10) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Adding two consecutive triangular numbers 6 and 10 gives: 1

(a) 18 **(b)** 15 **(c)** 16 **(d)** 20

2. The sum of the first 25 odd numbers is: 1

(a) 625 **(b)** 650 **(c)** 50 **(d)** 600

3. An angle of 90° is called a: 1

(a) obtuse angle **(b)** reflex angle **(c)** acute angle **(d)** right angle

4. A straight angle measures: 1

(a) 360° **(b)** 270° **(c)** 90° **(d)** 180°

5. In the Collatz process, if the number is odd we: 1

(a) multiply by 3 and add 1 **(b)** halve it
(c) subtract 3 **(d)** square it

6. One step of the Kaprekar routine on 8741 gives: 1

(a) 7173 **(b)** 7263 **(c)** 7272 **(d)** 7353

7. A bar graph has the categories written on the: 1

(a) axis along the base **(b)** title
(c) key **(d)** range

8. The difference between the largest and the smallest observation is the: 1

(a) key **(b)** scale **(c)** range **(d)** mean

9. The HCF of 18 and 30 is: 1

(a) 12 **(b)** 6 **(c)** 7 **(d)** 5

10. The largest 2-digit prime is: 1

(a) 98 **(b)** 97 **(c)** 91 **(d)** 99

11. The perimeter of an equilateral triangle of side 7 cm is: 1

(a) 49 cm **(b)** 21 cm **(c)** 14 cm **(d)** 28 cm

12. Area of a rectangle of length 25 cm and breadth 4 cm is: 1

(a) 125 cm^2 **(b)** 100 cm^2 **(c)** 58 cm^2 **(d)** 29 cm^2

13. Fractions with the same value are: 1

(a) unlike fractions **(b)** unit fractions
(c) equivalent fractions **(d)** proper fractions

14. A fraction whose numerator is greater than or equal to its denominator is a: 1

(a) proper fraction **(b)** improper fraction
(c) unit fraction **(d)** decimal

15. A rectangle of sides 9 cm and 5 cm has perimeter: 1

(a) 45 cm **(b)** 28 cm **(c)** 30 cm **(d)** 14 cm

16. A perpendicular is drawn from a point to a line to find: 1

(a) angle sum **(b)** the shortest distance
(c) the longest distance **(d)** perimeter

17. A figure has rotational symmetry if it: 1

(a) looks the same after rotating less than a full turn about a point
(b) has many colours
(c) has a centre of colour
(d) looks same after folding

18. Which letter has a horizontal line of symmetry? 1

(a) L **(b)** R **(c)** A **(d)** E

19. The value of $-6 + (-4)$ is: 1

(a) −8 **(b)** −12 **(c)** 10 **(d)** −10

20. The greater integer between −3 and −7 is: 1

(a) −7 **(b)** 0 **(c)** both equal **(d)** −3

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. Name two uses of patterns in everyday life. 2

22. What is the difference between the smallest and largest 3-digit palindromes? 2

23. Why should a graph have a title? 2

24. Why is 1 neither prime nor composite? 2

25. Find the area of a triangle with base 9 cm and height 4 cm. 2

26. What does the fraction $\frac{2}{3}$ mean when a rope is divided? 2

27. What is the use of a divider? 2

28. The temperature at night is −3°C and rises by 8°C by noon. What is it at noon? 2

SECTION — C (Short Answer Type) 8 × 3 = 24

Answer in about 30–40 words each.

29. Which is greater: the 7th square number or the 5th cube number? Show working. 3

30. What is the sum of angles on a straight line? If one angle is 110°, find the other. 3

31. What can we conclude from a bar graph where the bar for class VI is twice the bar of class V? 3

32. Convert 2.5 m² to cm² and 50,000 cm² to m². 3

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33. State the difference between a chord and a diameter. 3

34. Why does a scalene triangle not have a line of symmetry? 3

35. Write the first 8 multiples of 9. What do you notice about their digit sums? 3

OR

What is the smallest number that leaves remainder 1 when divided by 2, 3 and 4? 3

36. A submarine at −200 m rises 75 m, then dives 40 m. Where is it? 3

OR

The temperature of a city drops by 3°C each hour from 8°C for 5 hours. What is the final temperature? 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 80–100 words each.

37. (i) Find the angle between the clock hands at 5:00, 7:00 and 11:00. 5
(ii) Name each type.
(iii) How many degrees does the minute hand move in 25 minutes?

OR

(i) How do you compare two angles by tracing? 5
(ii) How do you compare by measuring?
(iii) Which is the more accurate way?

38. (i) Estimate 3,987 + 5,012 + 1,008. 5
(ii) Estimate 49 × 21.
(iii) Compare with exact values.

OR

(i) Write the Collatz rules. 5
(ii) Find the sequence for 11.
(iii) Count the steps.

39. **Case study:** A water tank is $\frac{5}{8}$ full. Rahul adds $\frac{1}{4}$ of the tank. Then $\frac{1}{8}$ is used. 5
(i) Fraction after adding?
(ii) Fraction after using?
(iii) Is the tank full at any time?
(iv) How much more is needed to fill?

OR

(i) Add $\frac{3}{4} + \frac{5}{6}$. 5
(ii) Subtract $\frac{7}{8} - \frac{1}{3}$.
(iii) Express both results as mixed fractions where needed.

40. (i) Define rotational symmetry. 5
(ii) Find the order and angle for a square, an equilateral triangle and a regular hexagon.
(iii) Which has the largest order?

OR

(i) How can you create a symmetrical pattern using paper folding? 5
(ii) Make a four-fold design.
(iii) Count its lines of symmetry.

END OF QUESTION PAPER — Total: 80 Marks

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